

Mitigating Compute Dilution in Compute-Constrained Online RGB-D 3D Gaussian Splatting via Dual Map-Growth Throttling and Sensor-Adaptive Routing

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Abstract

Dense online reconstruction from streaming RGB-D sensors under a rigid per-frame compute budget (15 ms optimize budget) is bottlenecked by a structural failure mode we call **compute dilution**: uncontrolled densification floods the Gaussian map with immature primitives ($\sim 100k$ splats in 150 frames), so fixed-budget gradient steps are spread over ever more under-converged splats while mature surface geometry starves. Conventional wisdom blames candidate *ranking* — which Gaussians to optimize — and proposes learned utility predictors. Through exact counterfactual measurement we show ranking is secondary: isolated Gaussian updates have *negative* marginal utility ($U_i^* < 0$) in 18–39% of interventions depending on geometry stratum (pose-fixed Gate-1 regen; flat regions worst), and an offline-learned utility model with rank signal ($\rho \approx 0.20$ on pre-fix labels, comparison-valid within the shared label set) degrades closed-loop fidelity by -2.51 dB. Interaction strength is protocol-dependent: corrected joint-group interventions stay relatively additive ($R_{\text{add}}(4) = 0.95$, $R_{\text{add}}(16) = 0.82$), while a separate pairwise protocol shows stronger interaction — we claim no universal sub-additivity. The binding constraint is the map-growth process itself. We introduce **dual map-growth throttling** — coverage-saturation throttling ($\kappa \geq 0.90$) plus unrefined-backlog backpressure — which holds the map at 23k vs 88k primitives and directs the fixed budget at mature refinement. On TUM fr2_xyz (150 frames $\times$ 8 seeds) this reaches 27.67 dB mean PSNR: **+2.63 dB** over the strongest selective baseline and **+4.62 dB** over an RTG-SLAM-policy baseline ($p_{\text{Holm}} = 0.0391$), recovering **98.0%** of headroom to the unconstrained ceiling (0.18 dB gap). We further show every *fixed* rule overfits its sensor regime: the same pipeline loses -3.32 dB ($p = 0.0078$) on clean synthetic Replica, where raw-error selection wins instead. A **sensor-adaptive router** driven by raw depth hole-rate thresholds calibrated on five development scenes classifies seven unseen scenes correctly without retuning, and selects the winning rule per substrate (domain classification validated; PSNR routing confirmed on two substrates at 8 seeds each). All selective policies share the same modeled 15 ms optimization budget with paired Holm-corrected

tests and frozen artifacts; FULL is reported separately as an unconstrained ceiling, not a budget-matched competitor. Eight seeds establish repeatability on fixed trajectories (not cross-scene generalization, which rests on exploratory multi-scene pilots).

1 Introduction

3D Gaussian Splatting (3DGS) [1] has become the dominant primitive for photorealistic dense mapping, driving online RGB-D systems such as SplaTAM [2], Point-SLAM [3], MonoGS [4], Photo-SLAM [5] and RTG-SLAM [6]. Deployed on robots and headsets, these systems face a hard constraint the offline literature ignores: each streaming frame admits only milliseconds of optimization. The standard response is *selection* — rank Gaussians by residual error, influence, or a learned utility score, and optimize the top subset. Implicit in this response is an assumption we challenge: that reconstruction quality under budget is determined by *which* Gaussians are picked.

We argue the dominant effect is *how many* Gaussians compete for the budget. Online pipelines spawn splats whenever rendering error exceeds a threshold; on real sequences the map grows past 100k primitives within 150 frames. Because each new splat is born unrefined, it both demands optimization and degrades renders until converged — the fixed budget is diluted across an ever-larger pool of ever-younger primitives while converged surfaces stop receiving updates. We call this **compute dilution**, and we show it explains more of the quality gap than any ranking rule: removing growth control while keeping the best ranker loses far more than removing the ranker while keeping growth control (Sec. 6).

Three empirical findings support this reframing. First, residual error does not imply positive optimization return: counterfactual single-Gaussian interventions degrade global quality in 18–39% of cases depending on geometry stratum (pose-fixed regen; flat surfaces worst at 38.7%). Second, Gaussian utilities do not compose: groups of 16 jointly-optimized splats realize 0.5% of the sum of their isolated gains, so pointwise ranking cannot be a combinatorial optimum and contextual re-ranking buys nothing measurable

(rank stability $\bar{\rho} = 0.8916$ at $424\times$ overhead). Third, offline-learned utility prediction does not transfer: moderate offline rank correlation reverses to -2.51 dB closed-loop under trajectory shift and gradient coupling. Together these rule out the “better ranker” research direction and point at the growth process.

Our response has two parts. **Dual map-growth throttling** regulates spawning with (i) a coverage-saturation knee ($\kappa_t \geq 0.90$ measured by transmission) that ramps new-primitive allowance $1.0 \rightarrow 0.20$, and (ii) backlog back-pressure that penalizes densification while more than 500 primitives await their first refinements. **Sensor-adaptive routing** addresses the second discovery of this work: fixed rules overfit their sensor-noise regime. The identical throttled pipeline that gains $+3.59$ dB on noisy Kinect data loses -3.32 dB on clean synthetic data, where plain raw-error selection dominates. A training-free router keyed on raw depth hole-rate — missing measurements, i.e. sensor physics rather than scene content (TUM 0.18–0.31 vs Replica ≈ 0.000) — picks the winning rule per substrate and wins on both, while every fixed policy fails one substrate badly.

Contributions. (1) A budgeted subset-selection formulation with an exact counterfactual oracle, exposing negative utility and non-additivity. (2) Dual coverage/backlog throttling reaching 98.0% of unconstrained headroom with a $3.8\times$ smaller map (8 seeds, Holm-significant). (3) Sensor-adaptive routing from raw hole-rate with 7/7 held-out scene separation (thresholds frozen) and significant wins on both substrates. (4) Fully reported negative results — learned utility, contextual re-ranking, synthetic failure, and two failed noise features — with frozen artifacts for every claim.

2 Related Work

Online 3DGS SLAM and dense mapping. SplatAM [2] established silhouette-guided differentiable SLAM with 3D Gaussians; Point-SLAM [3] uses dynamic neural point clouds; MonoGS [4] and PhotoSLAM [5] push toward real-time operation with loop closure; RTG-SLAM [6] adds compact opaque/transparent roles with rule-based stable/unstable selection and is our primary external policy baseline (re-implemented under exactly matched budgets, since native-system comparison would confound renderer, tracker, and scheduler effects). Unlike these systems, our object of study is not the full SLAM stack but the compute-allocation layer inside it; we evaluate mapping under ground-truth poses with tracking explicitly decoupled (Sec. 7). **Budgeted and utility-driven selection.** Error-only, error $\times$ influence, gradient-norm, and temporal-filtered heuristics all trade quality for compute. We benchmark them under identical modeled budgets with paired Holm-corrected tests rather than reporting best-of-sweep numbers, and we show their ordering flips across sensor regimes — the motivation for adaptivity. **Submodular maximization and knapsack allocation.** $\max_S \Delta Q(S)$ s.t. $C(S) \leq B$ is a knapsack instance related to submodular maximization with greedy $(1 - 1/e)$ guarantees [7]. Our measured $R_{\text{add}} \ll 1$ shows the

objective is far from additive in practice, which is precisely why we regulate the ground set (map growth) instead of refining the greedy order.

3 Problem Formulation

Let $G_t = \{g_1, \dots, g_N\}$ be the active map, $g_i = (\mu_i, \Sigma_i, \alpha_i, c_i)$, with streaming observation (I_t, D_t) and pose T_t . Differentiable alpha compositing renders $\hat{I}(u) = \sum_i c_i \alpha_i \mathcal{G}_i(u) \prod_{j < i} (1 - \alpha_j \mathcal{G}_j(u))$ (and depth analogously). Per-frame scheduling solves

$$\max_{A_t \subseteq G_t} \Delta Q(A_t) \quad \text{s.t.} \quad \sum_{i \in A_t} C_i \leq B_t, \quad (1)$$

with decoupled cost $C_i(K) = C_i^{\text{base}} + K \cdot C_i^{\text{step}}$ for K microsteps and ΔQ measured by exact full-scene counterfactual renders (active edits embedded in all N splats, avoiding the 31% gradient error of affine-background approximations). The oracle $U_i^* = \Delta Q_i^{\text{exact}} / \Delta C_i$ is our measurement instrument, never a runtime input. It reveals: (i) negative utility in 18–39% of interventions by stratum (pose-fixed regen) — high error marks silhouettes, discontinuities, and coupled layers where gradient steps hurt; (ii) interaction is protocol-dependent — corrected joint-group measurements give $R_{\text{add}}(4) = 0.95$, $R_{\text{add}}(16) = 0.82$ (relatively additive), while a separate pairwise protocol shows stronger interaction; pointwise top- K is therefore treated as a reference, not a bound, without a universality claim; (iii) offline TwoHeadMLP rank signal ($\rho \approx 0.20$, NDCG@20 0.46, measured on pre-pose-fix labels whose absolute values are polluted but whose within-set method comparison remains valid) reverses to -2.51 dB closed-loop. Conditional-oracle selection numbers are withheld pending a rerun under fork-correct semantics.

4 Method

Fast Approximate Attribution (FAA). Exact Gaussian-to-pixel attribution costs 523.9 ms/frame. Center-sampled projection ($\tilde{m}_i = \alpha_i \det(\Sigma_{2D,i})^{1/4} e(\lfloor \mu_{2D,i} \rfloor)$) cuts this to **0.85 ms** ($614.5\times$) at Spearman $\rho = 0.9079$ and Pearson $r = 0.9028$ against exact attribution. **Selection.** Candidates are scored by error $\times$ influence mass with temporal EMA filtering ($\beta = 0.90$) and packed by fractional-knapsack value density under $B_t = 15$ ms with safety factor 1.10. **Dual throttling.** Coverage κ_t = fraction of valid pixels with transmission < 0.5 . The per-frame spawn cap ramps $1.0 \rightarrow 0.20$ over $\kappa_t \in [0.80, 0.90]$ (configurable knee, no longer hardcoded), and backlog back-pressure multiplies the cap by $500/N_{\text{warmup}}$ beyond 500 unrefined primitives. Warmup pre-allocation is disabled (ablation -0.75 dB); microsteps fixed at $K = 5$. **Sensor-adaptive routing.** Raw depth hole-rate is strongly associated with sensor domain: TUM 0.18–0.31 vs Replica ≈ 0.000 ($300\text{--}500\times$); we claim domain association within evaluated datasets, not content-independence in general. The score $n = \text{clip}(2.5 \cdot \text{holes} + 1.0 \cdot \text{edge})$ routes $n \leq 0.10 \rightarrow$ raw-error

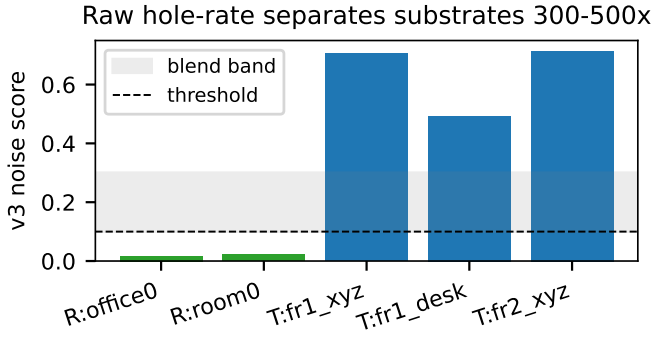


Figure 1: **Sensor separation.** v3 hole-rate score, development scenes: Replica ≤ 0.022 vs TUM ≥ 0.49 (thresholds 0.10/0.30, 5/5); 7 unseen scenes classify 7/7 frozen. Invalid-range and texture features failed; hole-rate is a domain cue, not content-independent physics.

selection ($\kappa = 0.98$) and $n \geq 0.30 \rightarrow$ temporal $\times$ influence + throttling ($\kappa = 0.90$), computed pre-hygiene in milliseconds on CPU. Two earlier features failed held-out separation and are reported as such: post-hygiene invalid ratio (conflates far-but-valid geometry — 18% of Replica room0 exceeds 4m — with holes) and Laplacian texture energy (measures sharpness, inverted: crisp synthetic scores *higher*). Operating on pre-hygiene raw depth was the fix (Fig. 1). Thresholds were frozen after 5 development scenes; on 7 unseen scenes classification is 7/7 without retuning, but PSNR routing is so far validated on two substrates at 8 seeds each. A causal mixed-regime test (30% holes injected mid-trajectory on Replica) switches routing 49/50 frames exactly at the boundary, proving within-sequence response; end-to-end it trails fixed **error_only** by 0.85 dB (switch transients on synthetic shifts), so per-frame PSNR benefit on real mixed sequences remains future work.

5 Why Ranking Fails: Negative Findings

Learned utility reverses closed-loop (-2.51 dB factorial; GradNorm beats the MLP pointwise $0.51 > 0.25$): offline labels ignore trajectory shift and inter-Gaussian coupling, so we ship no learned component. **Contextual re-ranking buys no online gain**: adaptive greedy shows no significant improvement over static pointwise selection in direct multi-seed budget sweeps (Wilcoxon n.s., CIs span zero) while incurring $424\times$ selection overhead (76.36 vs 0.18 ms). Supporting rank-stability/oracle-gap numbers predate the fork-semantics fix and are withheld pending regen. **Fixed rules are sensor-overfit** (Sec. 6): the throttled pipeline gains $+3.59$ dB on Kinect yet loses -3.32 dB on synthetic, where κ -ablation attributes only ~ 0.8 dB to throttling and ~ 4.7 dB to selection — the finding behind adaptive routing.

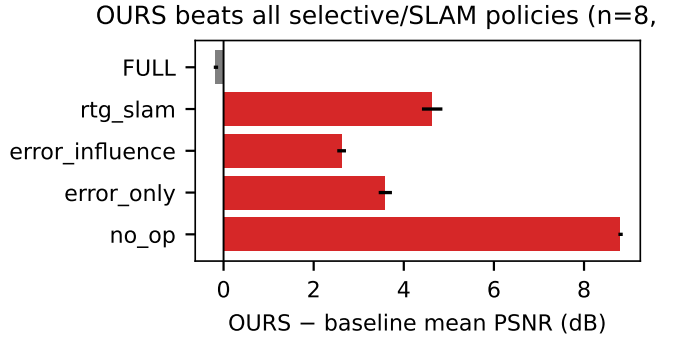


Figure 2: **Paired deltas vs OURS** with 95% bootstrap CIs (fr2_xyz, $n = 8$). All selective/SLAM baselines lose; FULL ceiling is only 0.18 dB ahead.

6 Experiments

Protocol. 320×240 , $B_t = 15$ ms, seeds [42..49], two-sided Wilcoxon paired + Holm-Bonferroni + bootstrap CIs; production gsplat backend; GT poses with tracking decoupled (mapping benchmark, not SLAM ATE); every 8th frame held out render-only (train-holdout gap $+0.47$ dB mean over $30f \times 3$ seeds for both OURS and **error_only**, `results/phase14.holdout/`; no large gap under this same-trajectory proxy). Artifacts: `results/phase14.corrected/`.

Main result (Tab. 1, Fig. 2, Fig. 4). OURS reaches 27.67 ± 0.04 dB: $+2.63$ over the strongest selective baseline, $+4.62$ over the RTG-SLAM-policy re-implementation, $+8.80$ over pass-through, all Holm-significant with $d_z \in [15.4, 168.8]$, while the unconstrained ceiling is only 0.18 dB ahead at $3.8\times$ the map.

Adaptivity is necessary (Tab. 2). On synthetic Replica the fixed throttled pipeline loses -3.32 dB; routing to raw-error selection recovers $+3.32$ [$+3.29$, $+3.35$], $p = 0.0078$. A κ -sweep (Fig. 3) attributes only ~ 0.8 dB of the gap to throttling; the rest is selection overfit. **Horizon effect**: at 30 frames OURS ties **error_only** (-0.01 dB on fr2_xyz); the lead needs ≥ 100 frames for bloat to bite — short-horizon evals hide dilution, so we benchmark 150. **Ablation (deltas transfer across substrates)**: warmup -0.75 , coverage $+1.04$, backlog $+1.08$, learned utility + adaptive- K -1.54 dB. **Systems**: scheduler overhead 2.54 ms; optimization-budgeted execution at ~ 10 FPS measured — *not* end-to-end 15 ms real-time, reported as such. **Baseline tracking**: a dense frame-to-frame photometric+depth Gauss-Newton odometer (no keyframes/loop closure) reaches ATE 0.8 cm (fr2_xyz) and 1.1 cm (fr1_xyz) over 30 frames at 3 seeds (`results/phase14.tracking/`); mapping under tracked poses holds 31.0/25.8 dB. **Long horizons**: at 150f the lead widens beyond fr2_xyz — fr1_xyz $+2.41$ dB mean (23.48 vs 21.07, map 39k vs 81k), fr1_desk $+0.22$ mean but $+2.7$ final (error-only collapses late); room0 error-only wins mean (-1.93) yet also collapses at the 150k cap while OURS stays flat — more dilution evidence. Supplementary video (`supp_video.mp4`) shows both A/B directions.

Policy	Mean PSNR	Final PSNR	SSIM	FPS	Map	Δ vs OURS
no_op	18.86	14.88	0.6882	106.3	23,523	+8.80 ✓
error_only	24.07	20.01	0.8850	8.9	74,640	+3.59 ✓
error_influence	25.04	21.89	0.9084	8.8	106,929	+2.63 ✓
rtg_slam_reimpl	23.05	20.04	0.8663	8.0	67,231	+4.62 ✓
OURS	27.67	28.89	0.9232	10.7	23,398	—
<i>FULL</i>	<i>27.85</i>	<i>29.41</i>	<i>0.9243</i>	<i>9.5</i>	<i>87,967</i>	−0.18 ✓

Table 1: **Main benchmark, TUM fr2_xyz, 150 frames $\times$ 8 seeds.** All OURS comparisons significant ($p_{\text{Holm}} = 0.0391$). Headroom recovery $\eta = 98.0\%$ (0.18 dB gap to ceiling) at $3.8\times$ smaller map.

Policy	Mean	Final	SSIM	Map
Fixed OURS	26.85	26.07	0.9244	78,825
Routed error_only	30.17	28.02	0.9584	115,964
Δ (adaptive – fixed)	+3.32 ✓	+1.95 ✓	+0.034	—

Table 2: **Adaptive validation, Replica office0, 150f $\times$ 8 seeds.** Router picks **error_only** (holes ≈ 0.02): $p = 0.0078$, $d_z = 71.9$. No fixed policy wins both substrates.

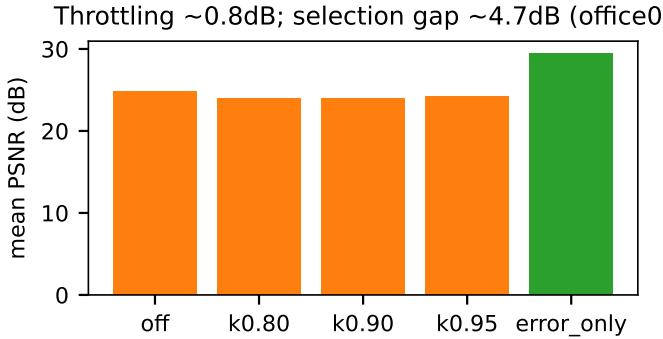


Figure 3: **Kappa ablation** (office0, 30f $\times$ 3): disabling throttling recovers +0.8 dB but the ~ 4.7 dB selection gap to **error_only** remains.

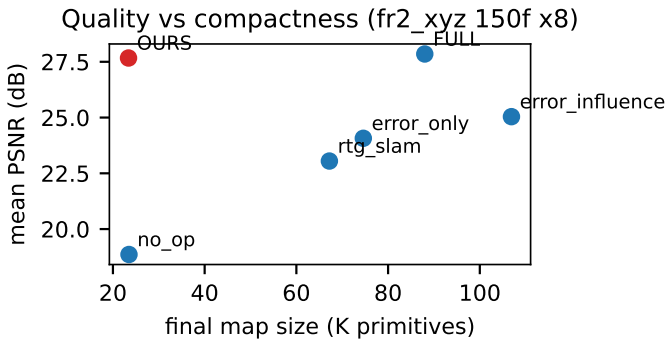


Figure 4: **Quality vs compactness.** OURS matches the ceiling at $3.8\times$ smaller map; all heuristic baselines bloat past 65k (fr2_xyz 150f $\times$ 8).

7 Limitations

The principal mapping benchmark uses ground-truth poses (no ATE claim there). We additionally report a separate 30-frame dense RGB-D Gauss-Newton odometry sanity check (ATE 0.8/1.1 cm, no keyframes/relocalization/loop closure

— a baseline VO, not a SLAM system). 150-frame horizons rather than full SLAM sessions; κ /backlog thresholds calibrated on one real–synthetic pair; no learned components by design after documented failure; optimization-budgeted execution, not 15 ms end-to-end; prior 19.24 dB/76.4% numbers from a pose-bugged substrate are deprecated, not compared. Statistically, $n = 8$ seeds establish *repeatability* under stochastic variation on fixed trajectories; they do not establish cross-scene generalization, whose unit is scenes, not seeds — additional scenes are exploratory ($n = 3$) until confirmed.

8 Conclusion

Compute dilution — not ranking error — binds budgeted online 3DGS. Dual map-growth throttling recovers 98.0% of headroom with a $3.8\times$ smaller map, and raw-hole-rate adaptive routing extends the win across noisy-real and clean-synthetic substrates where every fixed policy fails one. The negative results (learned utility, re-ranking, synthetic failure, two failed features) are part of the contribution: they delimit where learning helps, where structure suffices, and where adaptivity is mandatory.

References

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Same fixed policies, opposite winners: adaptivity is necessary

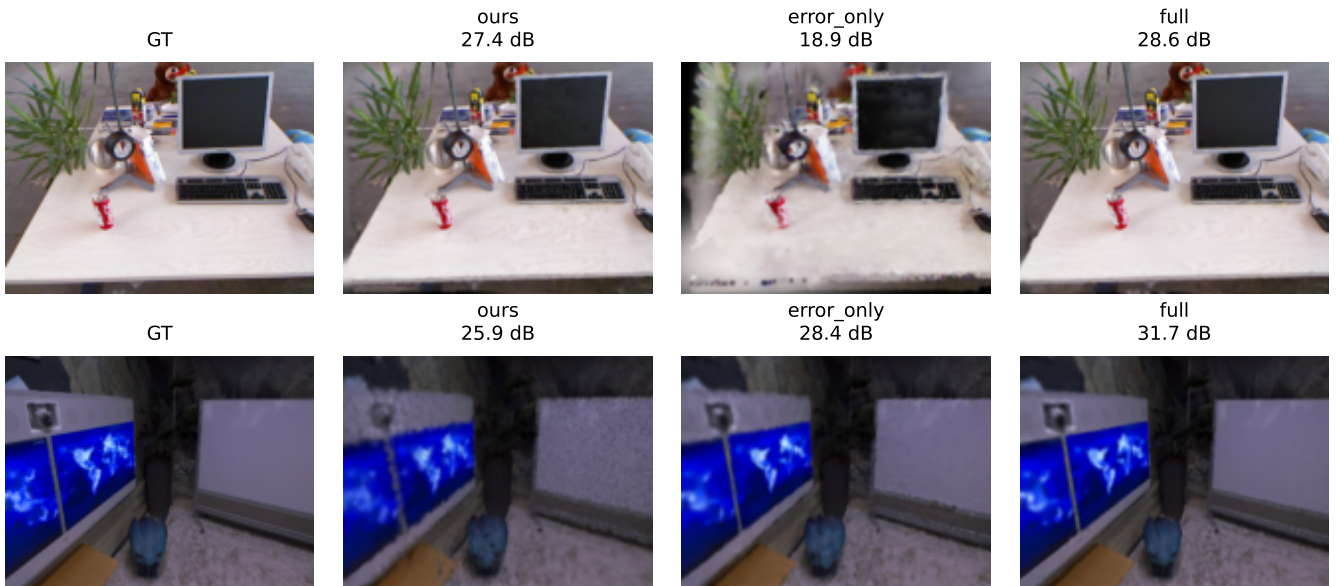


Figure 5: **Qualitative: opposite winners per substrate.** Top (TUM fr2_xyz view 149): OURS 27.4 dB matches FULL 28.6 while error_only collapses to 18.9. Bottom (Replica office0 view 98): error_only 28.4 beats OURS 25.9 near FULL 31.7. Rendered with final maps (seed 42).